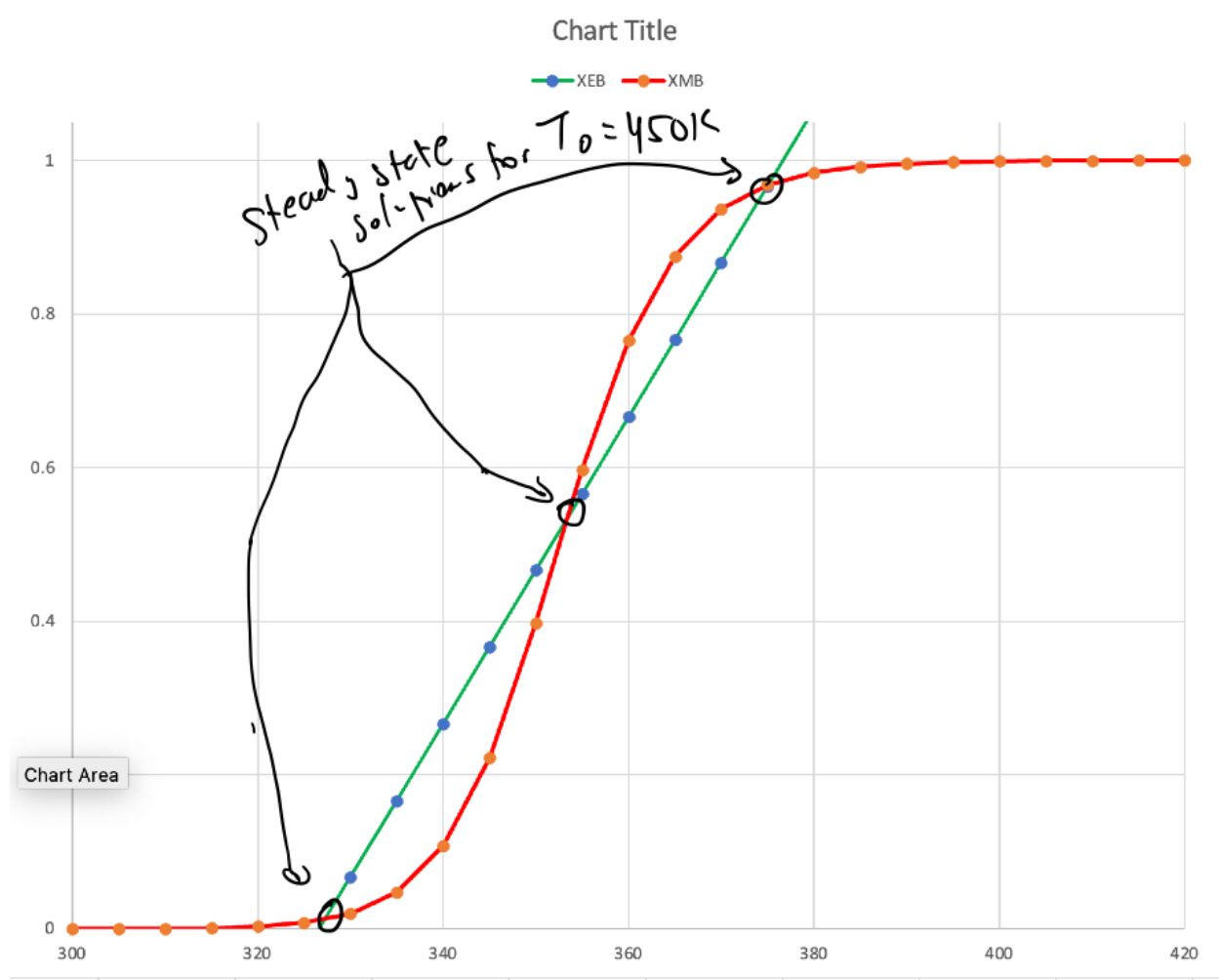


19 Steady state stability



- The steady state temperatures are the temperatures where $X_{EB} = X_{MB}$, i.e., where the X_{EB} & X_{MB} curves intersect
- The temperatures on the x-axis are reactor temperatures
- There can be multiple points of intersection for one feed temperature. For example, the feed temperature above has 3 steady state solutions.

- Steady state stability depends on $G(T)$ & $R(T)$ for T just slightly greater than that at the intersection point
- If $G(T) > R(T)$ the solution is unstable. This is b/c if the reactor temperature increases slightly, the heat exchanger can't quench it
- If $R(T) > G(T)$ the solution is stable, since the small increase in T will bring the reactor temp back down to this steady state
- Hence, in the plot above:

○ Reactor temperature = 324 K: $R(T) > G(T)$ for T just greater than 324 K, so this ss is stable

○ Reactor temperature = 354 K: $G(T) > R(T)$ for T just above 354 K, so this ss is unstable

○ Reactor temperature = 380 K:

$R(T) > G(T)$ for T just greater than 380 K, so this ss is stable

